EQUITY RESEARCH · INITIATING COVERAGE · FEBRUARY 2025

BUY

DUOLINGO, INC.

NASDAQ: DUOL · Technology — Education Software

Price Target $164.50 · Bull / Base / Bear: $165 / $97 / $60 · WACC: 11.1%

EQUITY RESEARCH · INITIATING COVERAGE

$164.50	$7.3B	11.1%	35.8%	49.62mm
Price Target (Bull DCF)	Enterprise Value	WACC	Revenue CAGR (FY20–24)	Fully Diluted Shares

EXECUTIVE SUMMARY

$164.50
Price Target (Bull)
DCF 5-year, diluted

$7.3B
Enterprise Value
PV FCFs + Terminal Value

11.1%
WACC
Rf 4.05% · β 1.59x · ERP 4.46%

$877.6mm
Net Cash
Zero debt; equity-only

BUY
Rating
Coverage initiation · Feb 2025

FINANCIAL SNAPSHOT ($mm, FY ends Dec 31)

Metric	2022A	2023A	2024A	2025E	2026E	2027E	2028E	2029E
Revenue ($mm)	369	531	748	1,010	1,313	1,667	2,067	2,522
YoY Growth	—	43.7%	40.8%	35.0%	30.0%	27.0%	24.0%	22.0%
Gross Profit	271	391	550	752	985	1,259	1,567	1,917
Gross Margin	73.3%	73.6%	73.5%	74.5%	75.0%	75.5%	75.8%	76.0%
EBIT ($mm)	(65)	(13)	64	111	170	249	347	466
EBIT Margin	(17.5%)	(2.4%)	8.5%	11.0%	13.0%	15.0%	16.8%	18.5%
Net Income ($mm)	(60)	16	89	126	184	262	357	473
EPS (Diluted)	**(1.21)**	**0.32**	**1.78**	**2.54**	**3.71**	**5.28**	**7.20**	**9.53**
FCF ($mm)	44	140	264	328	429	565	724	912

BULL CASE
$165 (25% prob.)

BASE CASE
$97 (50% prob.)

BEAR CASE
$60 (25% prob.)

PROB.-WTD.
$105 (Expected value)

01

Network-Effect Moat With AI Supercharger

Duolingo's 117mm MAU (37% CAGR FY20–24) creates a self-reinforcing data flywheel: more learners → better AI models → higher engagement → more learners. The Duolingo Max tier (powered by GPT-4) introduces roleplay and conversational practice at a $33/month premium — structurally expanding ARPU ceiling while driving paid conversion. No legacy edtech competitor can replicate this simultaneously at scale.

KEY DATA POINTS

— 117mm MAU (FY24); DAU/MAU improving to 31.6% — best-in-class engagement

— 12.1mm paid subscribers — 10.3% conversion (vs <5% in FY20)

— Duolingo Max (AI tier) priced at ~$33/month — 2× base Duolingo Plus

— Gamification + streaks create daily habit loop competitors can't replicate

02

Operating Leverage Just Beginning to Manifest

After years of heavy investment, Duolingo crossed EBIT profitability in FY24 ($64mm, 8.5% margin) from a -17.5% EBIT margin in FY22. We model EBIT margins expanding from 11% (FY25E) to 18.5% (FY29E) as revenue scales against a semi-fixed R&D and SG&A cost base. SG&A as % of revenue compressed from 53% (FY21) to 32% (FY24) — we see further 3–5pp of room.

KEY DATA POINTS

— EBIT positive for first time in FY24 (+8.5%); inflection point crossed

— Gross margin 73.5% → 76.0% (FY29E) as AI reduces human content cost

— R&D % declining ~100bps/yr as platform matures and engineers leverage AI

— SG&A % compression: word-of-mouth growth reduces paid acquisition costs

03

Underpenetrated Global TAM With Secular Tailwinds

Language learning is a $60B+ global market — Duolingo has <1% penetration. International expansion (particularly Asia-Pacific, LATAM, and MENA) provides a multi-year runway as smartphone penetration grows. New verticals (Math, Music, Chess) are early innings but structurally expand the addressable market by 20–30%. AI-native competitors (ChatGPT, YouTube) have not demonstrably dented retention metrics.

KEY DATA POINTS

— TAM: $60B+ global language learning; Duolingo penetration <1%

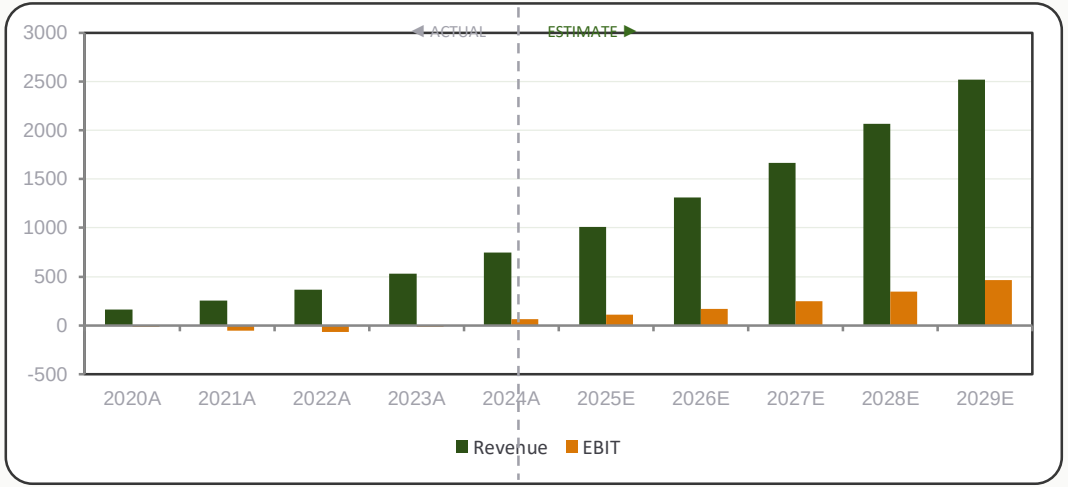
— Asia-Pacific: fastest-growing DAU cohort with 2× monetization upside

— Math + Music verticals: incremental TAM without new acquisition cost

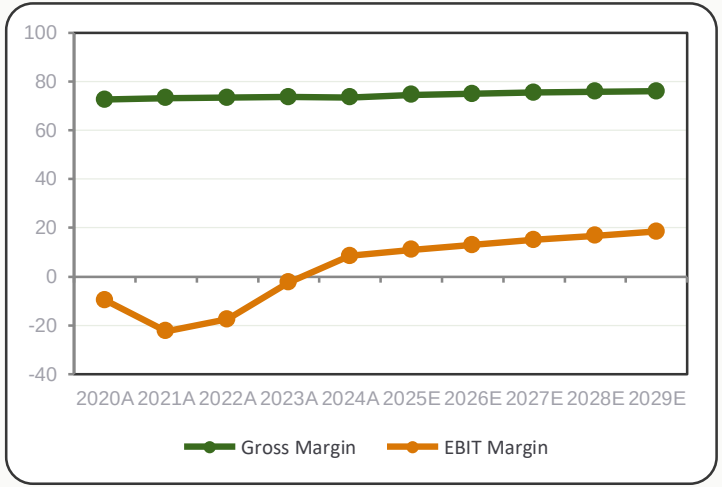
— No-debt balance sheet ($878mm cash); FCF generation accelerating sharply

FINANCIAL PERFORMANCE | FY2020A–FY2029E (BULL CASE)

REVENUE & EBIT ($mm)



MARGIN EVOLUTION (%)



KEY OPERATING METRICS

35.8%
Revenue CAGR
FY20–FY24 actual

24.5%
FCF / EV Ratio
PV FCFs vs EV

73.5%
Gross Margin
FY24A — software-like

12.1mm
Paid Subscribers
FY24A; +59% YoY

$878mm
Cash Position
Zero debt — clean BS

OPERATING METRICS | USER GROWTH & MONETISATION DRIVERS

USER GROWTH METRICS

Metric	FY20A	FY21A	FY22A	FY23A	FY24A	Comment
MAU (millions)	37	42	60	88	117	33% CAGR FY20–24
YoY MAU Growth	—	13.5%	42.9%	46.7%	33.0%	Consistent growth
DAU (millions)	9	11	16	27	37	Faster than MAU
DAU / MAU Ratio	24.3%	26.2%	26.7%	30.7%	31.6%	Improving engagement
Paid Subscribers (M)	1.8	2.9	4.8	7.6	12.1	61% CAGR
Paid Conversion %	4.9%	6.9%	8.0%	8.6%	10.3%	Key revenue driver
ARPPU ($/yr)	**$90**	**$86**	**$77**	**$70**	**$62**	**Mix-shift headwind**

Revenue = MAU × Paid Conversion × ARPPU

Growth driven by MAU expansion + conversion rate improvement, partially offset by ARPPU decline (international mix shift to lower-priced markets). Net effect: 40%+ top-line growth.

REVENUE PROJECTIONS BY SCENARIO ($mm)

Scenario	Revenue Growth Avg.	2024A	2025E	2026E	2027E	2028E	2029E
Bull Case (+27.6% avg)	**35%→22%**	**$748**	**$1,010**	**$1,313**	**$1,667**	**$2,067**	**$2,522**
Base Case (+21.4% avg)	28%→16%	$748	$957	$1,187	$1,437	$1,695	$1,963
Bear Case (+14.6% avg)	20%→10%	$748	$898	$1,050	$1,197	$1,341	$1,475

DCF VALUATION | 5-YEAR UNLEVERED FREE CASH FLOW MODEL

Duolingo, Inc. (DUOL) · NASDAQ · Technology — Education Software

WACC DERIVATION (CAPM — 100% EQUITY)

Risk-Free Rate (Rf)	4.05%
Equity Risk Premium (ERP)	4.46%
Levered Beta	1.59x
Cost of Equity (Ke)	11.14%
Weight of Equity	100%
Weight of Debt	0%
WACC	**11.14%**
Terminal Growth Rate	**2.50%**

UNLEVERED FREE CASH FLOW SCHEDULE ($mm)

FCF Build	2025E	2026E	2027E	2028E	2029E
EBIT	111	170	249	347	466
(–) Taxes on EBIT	(17)	(29)	(45)	(66)	(93)
NOPAT	94	141	205	281	373
(+) D&A	15	20	25	31	38
(–) CapEx	(26)	(34)	(44)	(54)	(66)
(–) Δ NWC	81	87	103	118	136
(+) SBC (non-cash)	106	138	176	218	266
Unlevered FCF	**270**	**352**	**464**	**593**	**746**
Discount Factor	0.949	0.853	0.768	0.691	0.622

EQUITY VALUE BRIDGE ($mm)

Sum of PV of UFCFs	$1,786mm	TV/EV: 24.5%
PV of Terminal Value	$5,500mm	TV/EV: 75.5%
Enterprise Value	$7,286mm	
(+) Cash & Equivalents	$878mm	
(–) Total Debt	$0	
Equity Value	**$8,164mm**	
FD Shares Outstanding	**49.62mm**	
Implied Price (Diluted)	**$164.53**	

SCENARIO ANALYSIS | BULL / BASE / BEAR CASE

BULL — Probability: 25%

$165

Implied Share Price (Diluted)

EV:	$7.3B
Rev. FY29E:	$2,522mm
EBIT Mgn FY29E:	18.5%
Rev. CAGR:	27.6%

BASE — Probability: 50%

$97

Implied Share Price (Diluted)

EV:	$3.9B
Rev. FY29E:	$1,963mm
EBIT Mgn FY29E:	11.3%
Rev. CAGR:	21.4%

BEAR — Probability: 25%

$60

Implied Share Price (Diluted)

EV:	$2.1B
Rev. FY29E:	$1,475mm
EBIT Mgn FY29E:	1.5%
Rev. CAGR:	14.6%

KEY ASSUMPTIONS BY SCENARIO

Assumption	Bull Case	Base Case	Bear Case
5-Year Revenue CAGR	27.6%	21.4%	14.6%
Terminal EBIT Margin (FY29E)	18.5%	11.3%	1.5%
R&D % of Revenue (FY29E)	26.1%	28.6%	33.0%
SG&A % of Revenue (FY29E)	29.9%	32.4%	35.0%
Gross Margin (FY29E)	76.0%	74.5%	71.0%
Expected Value (Probability-Weighted):		**$105 / share**	

SENSITIVITY ANALYSIS | WACC & EBIT MARGIN SCENARIOS

★ Bull Case ($164.53)

Above current price ($112)

WACC vs. TERMINAL GROWTH RATE : IMPLIED DILUTED SHARE PRICE ($)

WACC \ TGR →	1.50%	2.00%	2.50%	3.00%	3.50%
9.14%	$190	$200	$212	$226	$242
10.14%	$169	$177	$185	$195	$206
11.14% ◀ Bull	$152	$158	**$165**	$172	$180
12.14%	$138	$143	$148	$154	$160
13.14%	$127	$131	$135	$140	$145

TERMINAL EBIT MARGIN vs. REVENUE CAGR : IMPLIED DILUTED SHARE PRICE ($)

Margin \ CAGR →	20%	24%	27.5%	31%	35%
12%	$121	$133	$145	$158	$175
14%	$126	$138	$151	$165	$183
16%	$130	$144	$157	$172	$191
18.5% ◀ Bull	$136	$150	**$165**	$181	$201
21%	$141	$157	$172	$189	$211
23%	$146	$162	$178	$196	$219
25%	$150	$167	$184	$203	$227

RISKS | INVESTMENT CONCLUSION

KEY RISKS TO OUR BULL THESIS

AI Substitution Risk

Free AI tools (ChatGPT voice mode, Google Translate, YouTube immersive) could reduce the perceived need for structured language learning. While Duolingo leverages AI itself, a commodity shift in conversation practice would compress ARPU ceiling.

ARPPU Structural Decline

ARPPU has fallen from $90/yr (FY20) to $62/yr (FY24) as international growth dominates. If this decline accelerates or if Duolingo Max fails to upsell existing subscribers, revenue growth could undershoot even the base case.

Growth-to-Value Valuation Risk

At current prices (~$112), DUOL trades at ~27× FY25E EBITDA. Any material miss on subscriber growth or margin trajectory risks significant multiple compression given the high-growth premium embedded in the valuation.

Execution in New Verticals

Math and Music are unproven verticals. If they fail to attract paid subscribers, they represent sunk R&D cost with no revenue offset, compressing margins without the anticipated TAM expansion benefit.

INVESTMENT CONCLUSION

BUY

INITIATING COVERAGE

Price Target: $164.50 · Bull DCF · +47% Upside

Duolingo is the rare high-growth tech asset with a clear path to compounding profitability. The combination of network-effect moat, AI-driven product innovation, and underpenetrated global TAM justifies a premium multiple. We initiate at BUY with a $164.50 bull-case DCF target; probability-weighted expected value of $105 still represents material upside from ~$112 current levels.

NEAR-TERM CATALYSTS

- Q1 2025 earnings: subscriber net adds vs. consensus
- Duolingo Max global rollout & ARPU uplift tracking
- Math / Music vertical DAU + paid conversion metrics
- International (APAC) MAU acceleration vs. guidance

DUOLINGO, INC.

NASDAQ: DUOL · Equity Research · Initiating Coverage · BUY · PT: $164.50

Rating & Valuation

Rating	**BUY**
Price Target	**$164.50**
Expected Value	**$105 (prob.-wtd.)**
Enterprise Value	**$7,286mm**
Equity Value	**$8,164mm**
FD Shares	**49.62mm**

Key FY29E Estimates (Bull)

Revenue	**$2,522mm**
Revenue CAGR (FY24–29)	**27.6%**
Gross Margin	**76.0%**
EBIT	**$466mm**
EBIT Margin	**18.5%**
FCF	**$912mm**

DCF & Scenario

WACC	**11.14%**
Terminal Growth Rate	**2.5%**
Bull Case	**$165**
Base Case	**$97**
Bear Case	**$60**
Prob.-Wtd.	**$105**